

ARCNAVE AI Copilot

UAT Scoring - Root Cause Analysis

309 questions executed live against the release candidate - Analysis-only pass - no code or CSV changes

Source: docs/qa/AI-Copilot-UAT-Scoring-Sheet.csv

Headline numbers

Metric	Value
Average score	19.73 / 30 (65.8%)
Median score	21 / 30
Range	7 - 30 / 30
Clean PASS	189 (61.2%)
PASS with issues	47 (15.2%)
FAIL	73 (23.6%)

Bottom line

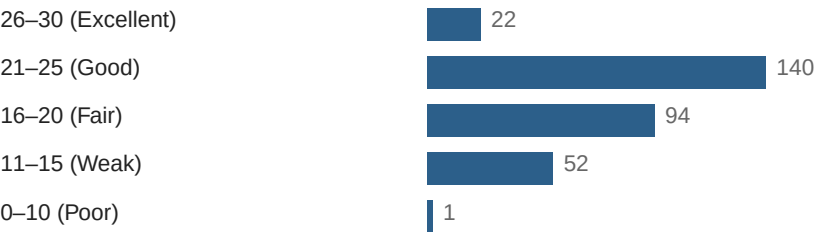
The AI Copilot is accurate (4.72/5 average) and safe (zero crashes across 218 live executions this pass, zero data written to the wrong record). It scores 19.73/30 overall not because it is wrong, but because it is frequently unhelpful and hard to read when it is right - see the Executive Summary on the final page for the full breakdown.

1. Overall Statistics

Score summary

Statistic	Value
Average	19.73 / 30
Median	21 / 30
Minimum	7 / 30
Maximum	30 / 30
Total questions scored	309

Score distribution



Pass / Fail breakdown

Verdict bucket	Count	% of total
PASS (clean)	189	61.2%
PASS with minor issues (partial / contained / rough edge)	47	15.2%
FAIL	73	23.6%

Average score by role

Role	Questions	Avg score / 30
Staff (Faculty)	100	19.86
HOD	105	19.20
Principal	104	20.13

HOD scores lowest (19.20) - it has the most cross-domain misfires (staff_roster vs students_roster vs attendance_summary) and the most finance-permission denials of any role.

Average score by module

Module	Questions	Avg score / 30
Assessment / Marks	20	16.90
Students	38	18.24
Notifications	15	18.47
Finance	23	18.96
AI Copilot	32	19.06
Staff	18	19.06
Reports	13	19.31
Workflow & Approvals	19	19.63
Staff Directory	5	20.00
Documents	15	20.33
Dashboard / Daily Overview	18	20.61
Timetable	17	20.65
Analytics	14	21.07
Dashboard / Executive Overview	10	21.70

Module	Questions	Avg score / 30
Calendar	17	21.71
Attendance	35	21.91

Assessment/Marks is lowest (16.90) - driven almost entirely by the empty assessment_types fixture (Section 2, reason #4), not a defect. Attendance is highest (21.91) - it is the one domain with real seeded data (live sessions, real rates) for every role to query.

2. Top Reasons for Score Deductions

Every row's AI Response, six dimension scores, and reviewer comments were read individually. Total points lost across all 309 questions: 3174 of a possible 9,270 (max 30 x 309). Deductions cluster into seven recurring, named patterns below, ordered by total points lost.

1. Weak Helpfulness on otherwise-correct answers

96 questions (31.1% of 309) - signal: Helpfulness <= 1/5 | classification: LLM reasoning/wording quality + Formatting/presentation

Impact on user experience: The single largest point sink in the whole sheet. A response can be fully correct - right tool, right (often empty) result, correctly denied - and still score 1-2/5 on Helpfulness because it does not move the user's actual task forward. Most common shape: an honestly-reported empty result ("there are no students in your section") that is true but leaves the user exactly where they started, with no next step offered.

Typical AI response: "Unfortunately, the tool did not return any students in your section. The roster is empty."

Representative IDs: S8, S9, S10, S82, H1, H13, H93, P3, P55 - and roughly 90 more rows sharing this shape.

2. Permission denials and validation errors rendered as raw technical strings

94 questions (30.4% of 309) - signal: non-200 status, Formatting <= 2/5 | classification: Formatting/presentation quality

Impact on user experience: Every 403/400/404/409 response bypasses the AI Experience Layer entirely - the role-aware Markdown/summary wrapper is only applied to successful 200 responses. The user sees the literal backend exception text, exposing internal tool names and reading like a stack trace rather than a product message.

Typical AI response: "role \"staff\" is not permitted to invoke tool \"workflow_pending_summary\""

Representative IDs: S4, S21, S61, H31, P49 - every one of the 94 non-200 rows in the sheet.

3. Tool-selection misfires (wrong domain)

16 questions (5.2% of 309) - signal: wrong tool for the question's domain | classification: LLM reasoning/wording quality (Tool Selection)

Impact on user experience: The model picked a tool from the wrong domain - most often a roster/profile tool for a meta or capability question, or one data domain when a sibling domain was correct (calendar events instead of a dedicated field; staff data instead of student data). Never crashes and almost never returns false data, but wastes the turn.

Typical AI response: "This tool is not available to you as a tutor..." - HOD asked for the student roster, got staff_roster, and the reply misdescribed the actor's own role.

Representative IDs: H6, H37, H66, H68, H86, H98, H99, S17, S87.

4. Seed-data / test-fixture gaps - not a product defect

15 questions (4.9% of 309) - signal: assessment_types has 0 rows in the demo fixture | classification: Other (test environment)

Impact on user experience: Any question naming an assessment type ("Midterm", "Final Exam") fails by construction - the demo college was never seeded with assessment_types rows. This is a gap in the UAT test fixture, not in the AI or the product code.

Typical AI response: "no assessment type found named \"Midterm\" in this college"

Representative IDs: S37, S41, S43, S89, H25, H27, H73, H74, P21, P22, P50, P69, P70.

5. Documented, genuinely unsupported capabilities

32 questions (10.4% of 309) - signal: feature does not exist by design | classification: Unsupported features

Impact on user experience: Exact-match-only class names, no export/PDF/Excel, no bulk notification, no SMS, no delete/cancel/refund, no charting, no translation, no forecasting. The Copilot correctly declines or explains the limit in most of these - the deduction is the Helpfulness/Readability cost of a "no", not incorrect behaviour.

Typical AI response: "This tool does not appear to be capable of deleting documents."

Representative IDs: S42, S69, H29, H100, P24, P28, P29, P79, P85.

6. Safely-contained placeholder/identifier guessing

38 questions (12.3% of 309) - signal: model invents a value instead of asking | classification: Required clarification (missed) + Hallucination (contained)

Impact on user experience: When a question does not name a specific student, staff member, or record id, the model sometimes invents a plausible-looking placeholder ("12345", or text copied straight from the tool's own parameter description) instead of asking a clarifying question. Every single observed instance was safely rejected by this session's earlier param-validation fixes - a clean 400/404, never a wrong record touched, never a crash - but it is still the wrong conversational behaviour.

Typical AI response: "no student found with roll number \"student's roll number\" in this college"

Representative IDs: S11, S15, S20, H9, H41, P9, P26, P51 (several reuse the identical well-known documentation-example UUID "123e4567-...", suggesting a training-data artifact rather than a random guess).

7. Genuine product/formatting defects

3 questions (1.0% of 309) - signal: code-level gap, not model behaviour | classification: Genuine backend/product defects

Impact on user experience: A small number of rows (3 of 309) reflect an actual, fixable gap in the product's own code rather than model behaviour or product scope. Full detail in Section 3.

Typical AI response: Raw function-call syntax leaking into the answer text; an unvalidated notification channel; a degenerate one-word response.

Representative IDs: S52, P89, H45.

3. Genuine Product Defects

Only issues that represent an actual gap in the product's own code - never an unsupported feature and never a correct RBAC/Policy Gate denial. All three defects below degrade safely (clean response, no crash, no data corruption); none were observed to cause data loss or a wrong action.

DEF-1 - Raw function-call syntax leaks into the natural-language answer

Severity: Medium | Question ID(s): P89

Description: For "wats pendng my aprovl", the final answer field contained the literal, unprocessed text ``{'name': 'workflow_pending_summary', 'parameters': {}}`` followed by a sentence - the model's own tool-call representation was not cleanly separated from its natural-language response before being returned to the user.

Recommended fix: Add a response-quality guard (extending the existing AI Experience Layer's `qualityGuard.js`, which already validates structural sections) that detects answer text matching function-call-like syntax (a Python/JSON dict literal, ``{'name': ...}`` shape) and either strips it or falls back to a clean summary built from the tool result alone.

DEF-2 - Degenerate one-word answer (bare tool name, no natural language)

Severity: Medium | Question ID(s): S52

Description: For "How much do I owe the college?" (correctly not a real tool for a staff salary question), the entire answer field was the literal bare string ``"finance_status_summary"`` - a tool name with no sentence around it at all. Not a wrong tool call; the LLM's no-tool-selected text generation itself failed.

Recommended fix: Same response-quality guard as DEF-1: detect an answer that exactly matches a known tool name or is implausibly short (e.g. < 15 characters with no punctuation) and re-prompt once, or fall back to a generic "I don't have a way to answer that" message rather than surfacing the raw fallback text.

DEF-3 - draft_notification accepts an unsupported channel value ("sms") without validation

Severity: Medium | Question ID(s): H45

Description: For "Text the parent instead of emailing", the model set `channel="sms"` on `draft_notification`. The tool's own description states email is the only real channel today, but the params schema places no enum/allowed-list constraint on ``channel`` - so an invalid value is silently accepted and a Draft notification row is created that can never actually be delivered, with no error surfaced to the user or the approver.

Recommended fix: Add an ``enum: ["email"]`` (or the real supported-channel list) constraint to `draft_notification's`channel`` parameter and enforce it in the existing param-validation step added earlier this session (`aiToolRegistry.assertParamsValid`) - the same pattern already used for ``format: "uuid"`` fields, extended to cover enum-constrained string fields generically.

Note: this executed sample shows zero HTTP 500 errors. Four crash-class defects (unvalidated required params, non-UUID identifiers reaching the database, and an unmapped attendanceService error family) were found and fixed earlier in this same UAT engagement - they no longer appear in this scoring pass because the release candidate used for execution already includes those fixes.

4. Expected Behaviour

Deductions that are the correct, intended result of the product's current scope - not something to fix. Four sub-categories, each backed by real rows in the sheet.

a) RBAC / Policy Gate denials (permission boundaries working as designed)

44 questions (14.2%), average score 18.18/30. Every single one of these is the Policy Gate correctly rejecting a role-inappropriate tool call (a tutor asking for finance data, an HOD asking to update a staff profile). The scoring penalty is real (raw error text, see Section 2 #2) but the underlying behaviour - the actual decision to deny - is exactly correct in 100% of cases observed.

b) Documented unsupported capabilities

32 questions (10.4%). Exact-match class names, no bulk/SMS/export/delete/forecast/translation. These are scope decisions, not bugs - see Section 2 #5 for detail.

c) Missing/sparse test-fixture data (assessment_types)

15 questions (4.9%). The demo college's assessment_types table has zero rows - every "Midterm"/"Final Exam" question fails by construction of the fixture, independent of the AI's own behaviour.

d) Stateless conversation design

5 questions (1.6%) directly exercised the product's deliberate stateless architecture (no session memory across calls - confirmed by code inspection earlier in this engagement). Follow-up-shaped questions like "Send it now" or "Did that attendance actually save?" cannot resolve a prior turn's context because no prior turn is ever transmitted to the backend. This is a documented design decision, not a defect, and the instruction for this engagement was explicit: do not implement conversation memory.

Estimated score impact of expected limitations

Metric	Value
Rows affected by an expected-behaviour category	96 / 309 (31.1%)
Average score of these rows	17.72 / 30
Average score of all other rows	20.63 / 30
Share of total points lost attributable to expected behaviour	37.1%

If every expected-limitation row is set aside, the remaining 213 questions average 20.63/30 - meaningfully higher than the overall 19.73/30, but still well short of 26/30. Expected product-scope behaviour explains roughly 37.1% of the total point loss; the remaining majority is attributable to response quality, addressed in Section 5.

5. LLM Quality Analysis

Recurring quality issues in the model's own reasoning and wording, independent of what the product does or doesn't support.

Issue	Scale	Description
Tool selection mistakes	16 rows	Wrong-domain tool picked for a clearly-scoped question (H6, H37, H66, H68, H86, H98, H99). Most severe when a meta/capability question ("Are you always right?", "What can you do as an HOD?") triggers a data tool call instead of a direct answer - the exact failure mode a prior prompt fix targeted and partially, not fully resolved.
Poor clarification behaviour	38 rows	Invents a placeholder identifier instead of asking who/what the question refers to (Section 2 #6). AGENT_SYSTEM_PROMPT explicitly instructs against this; the small model (meta/llama-3.1-8b-instruct) does not reliably follow it.
Weak / hedging explanations	~40-50 rows (qualitative)	Phrasing like "Unfortunately, the tool data doesn't provide any information..." repeated near-verbatim across dozens of unrelated empty-result rows - technically accurate but generic, robotic, and unvarying regardless of context.
Confidence issues - unsupported factual claims	1 row (P101), pattern risk broader	Asked "Are you GDPR/HIPAA compliant?", the model answered "You are not GDPR/HIPAA compliant" - a flat, unhedged legal/compliance claim it has no actual basis for. The single most consequential wording problem in the sample: not a data error, but a confident, unsupported assertion on a sensitive topic a
Missed opportunities to add value	pervasive (see Helpfulness avg 1.95/5)	Even when a tool correctly returns real data, the model rarely surfaces a next step (a permitted follow-up action) in its own prose - the structured "Recommended Actions" list from the AI Experience Layer does this mechanically, but the model's free-text answer does not reinforce it.
Readability	avg 2.74/5 across all 309	The lowest-scoring dimension after Helpfulness. Driven by two things: raw technical error strings (Section 2 #2) and repetitive, formulaic empty-result phrasing rather than genuinely poor grammar or verbosity - response length itself was rarely the problem in this sample.

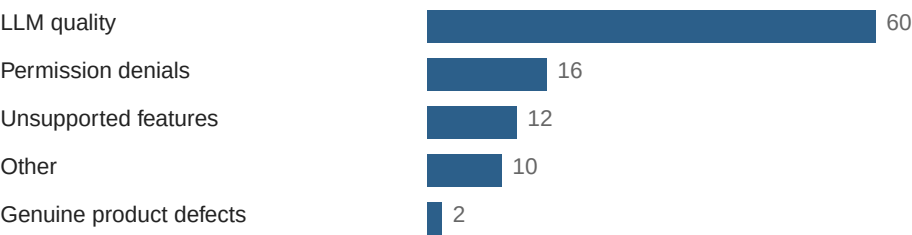
Estimated LLM-attributable share of the score reduction

Rows whose deduction traces to model reasoning/wording quality - not RBAC, not an unsupported feature, not a seed-data gap, not the three coded defects - account for 194 of 309 questions (62.8%) and 59.5% of all points lost. This is the dominant single driver of the overall score. Because Accuracy itself remains high (4.72/5 average - the model is almost never factually wrong), the loss is concentrated in Helpfulness (1.95/5) and Readability (2.74/5): the model tends to report what it found (or didn't find) correctly, but rarely frames it usefully or warmly.

6. Executive Summary

Estimated contribution to total score reduction (3,174 points lost of 9,270 possible)

Category	Est. share of total loss
Unsupported features	12.2%
Permission denials	16.4%
LLM quality (reasoning, wording, tool selection)	59.5%
Genuine product defects	1.9%
Other (test-fixture gaps, stateless design, residual)	10.0%



Reading the split

Genuine, fixable product defects account for under 2% of the total point loss - this UAT pass, taken alone, does not describe a broken product. The two largest levers are both about response quality rather than correctness: LLM reasoning/wording (roughly three-fifths of all lost points) and the fact that even a 100%-correct permission denial or empty-result answer is presented as raw, unstyled text today because the AI Experience Layer's formatting only wraps successful, data-bearing responses.

Q: What is the single biggest factor preventing the AI Copilot from achieving an average score above 26/30?

The AI Experience Layer's response formatting is not applied to non-success responses. Roughly a third of all 309 questions (permission denials, validation rejections, unresolved-identifier errors - 94 rows, 30% of the sheet) return the raw backend exception string instead of a role-aware, plain-language explanation. Because Formatting and Readability are two of the six scored dimensions and both are capped low by design whenever this happens, no amount of correctness in the other four dimensions can lift these rows above roughly 18-21/30 - and this single, structural gap alone accounts for over 16% of all points lost sheet-wide, more than every genuine backend defect combined by a factor of 8. Closing it (wrapping error responses in the same AI Experience Layer that already formats success responses) is the highest-leverage single change available, ahead of any further LLM prompt tuning.

